

KNIME



- **Modern data analytics platform that enables sophisticated statistical and data mining analysis on small, medium or large data**
- **Visual workbench chains data access, data transformation, initial investigation, powerful predictive analytics and visualization**
- **Provides the ability to develop reports or automate the deployment of new insights back into production systems**
- **KNIME Analytics Platform is open source and available under GPL license**
- **It can be extended with Commercial Software to include professional support, productivity and collaboration functionality**

KNIME Website Links

Web Links

http://www.knime.org	This is the first place to look for information about KNIME products. The open source Desktop version can be downloaded here.
http://www.knime.org/learning-hub	This is the landing page to learn more specific KNIME functionalities. Learning material - as web sites, videos, webinars, courses, and more - is organized by topic, like text mining or chemistry, or basic KNIME functionalities, etc...
http://tech.knime.org/forum	In the www.knime.org site you can find a number of resources. What I find particularly useful is the KNIME Forum. Here you can ask questions about how to use KNIME or about how to extend KNIME with new nodes. Someone from the KNIME community answers always and quickly.
http://tech.knime.org/knime-labs	This site contains nodes still under development; i.e. the beta version of new nodes. You can already download them and use them, but they are not of product/release quality yet.
http://knime.org/supporters-0	This is the site where all contributing supporters (partners, providers, and sponsors) are listed.

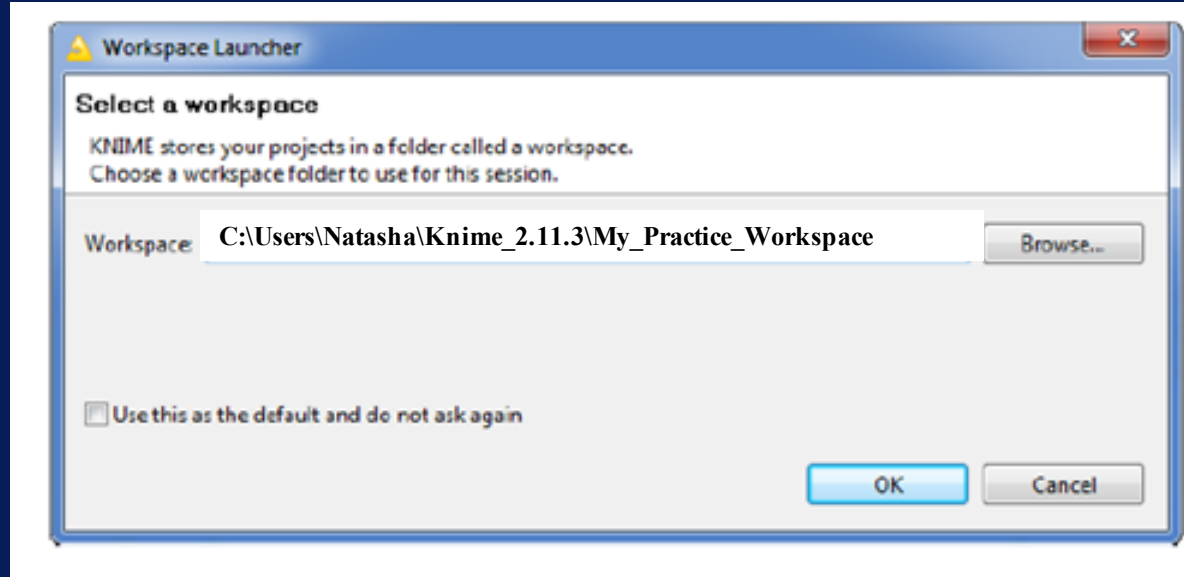
Download & Install KNIME

1. Go to: <https://www.knime.org/downloads/overview>
2. Select the “**Download KNIME**” tab.
 - You don't need to complete the Register for Help & Updates form.
3. Selected one of the following links to download **KNIME Analytics Platform**:
 - Windows: Select the **32-bit** or **64-bit** link to the right of **KNIME + all free extensions for Windows (installer)**
 - Mac: Select the **64-bit** link to the right of **KNIME + all free extensions for Mac OSX (10.7 and above)**
4. Accept the Terms and Conditions and select the **Download** button.
5. Install KNIME once the download is complete:
 - Windows: Double click the KNIME installer exe file and complete the install wizard.
 - Mac: Double click the KNIME dmg file then drag the KNIME app file to the Applications folder.

KNIME Workspace

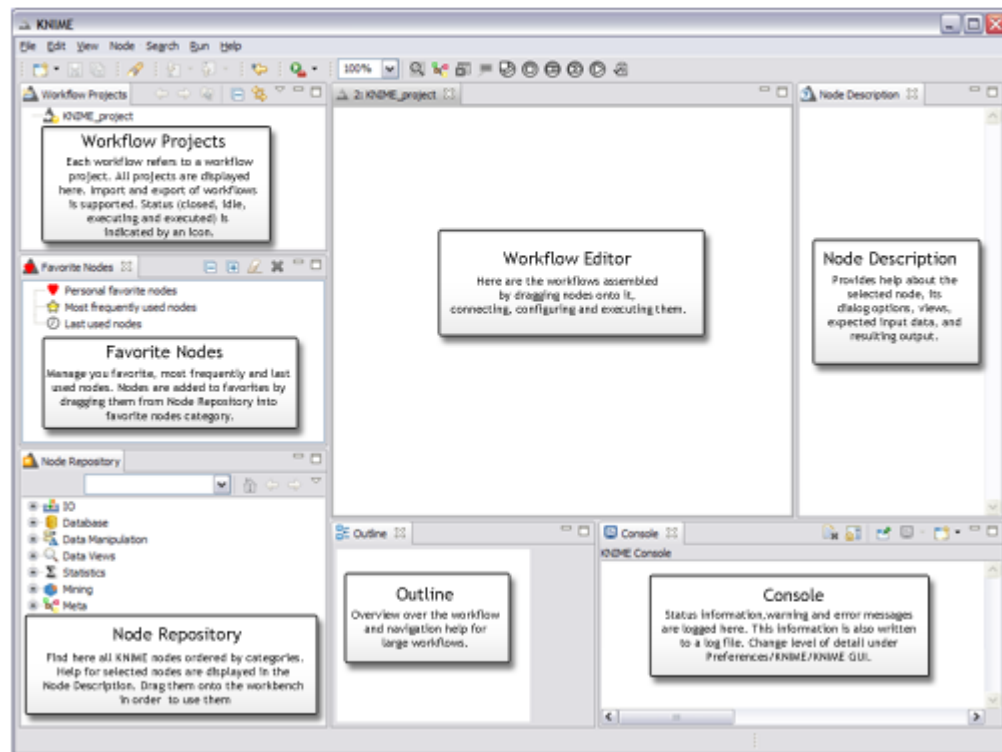
The Workspace is the directory where all current workflows and preferences are saved

It can be located anywhere (you specify) on the hard drive



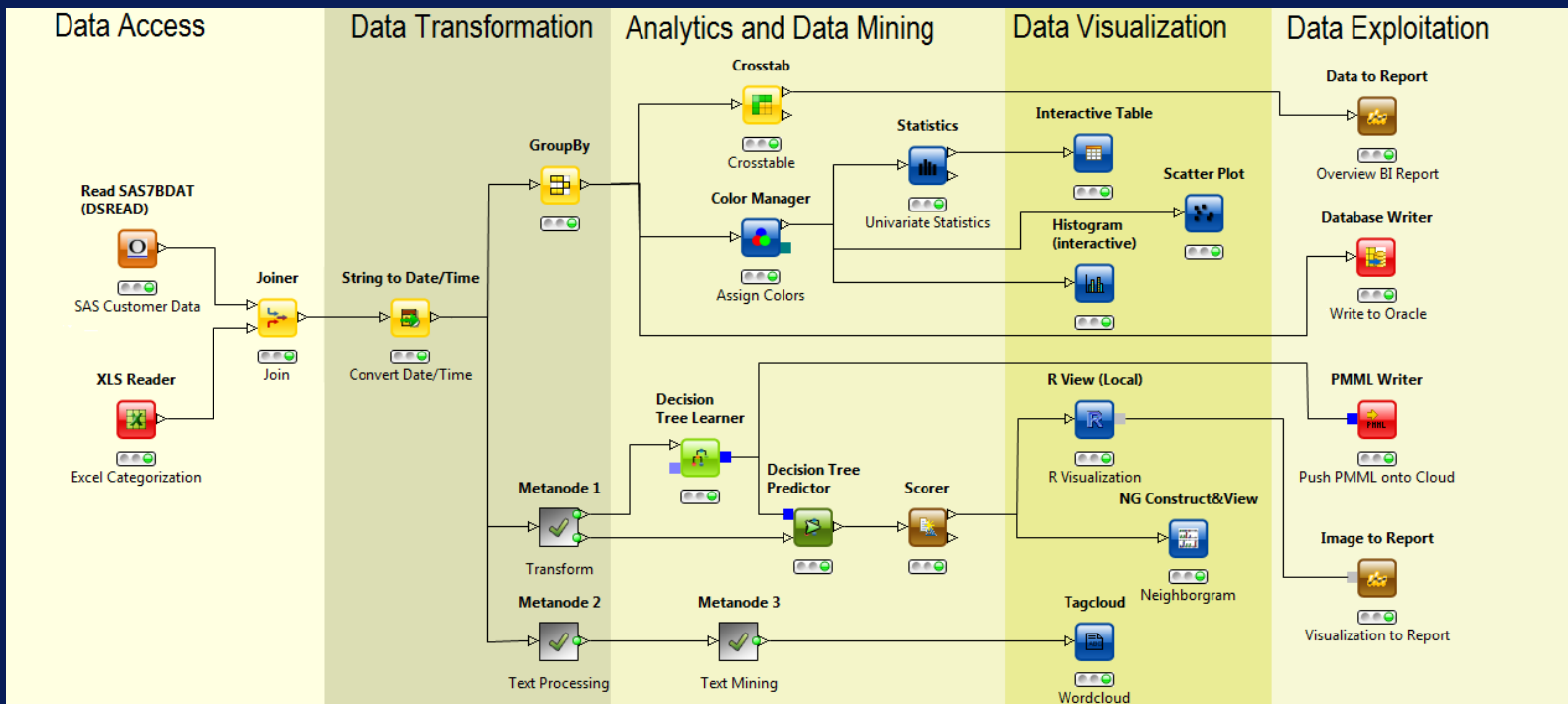
Workbench overview

The KNIME Workbench is organized as follows:



The image shows the KNIME v2.4.1.0030379 desktop application. The interface is divided into several panes:

- Workflow Projects:** A tree view on the top left showing a project hierarchy with folders like 'Chapter2', 'Exercises', and 'my_first_workflow'.
- Favorite Nodes:** A section below the projects showing 'Personal favorite nodes', 'Most frequently used nodes', and 'Last used nodes'.
- Node Repository:** A large pane on the bottom left containing a searchable list of nodes categorized by type (e.g., IO, Database, Data Manipulation, Statistics, Mining, Chemistry, Distance Matrix, Meta, Flow Control, Misc, KNIME Labs, Time Series, Quick Form, R, Reporting, Weka, XML).
- Workflow Editor:** The central workspace showing a workflow named 'my_first_workflow'. It contains a sequence of nodes: 'File Reader' (reading 'adult.data set'), 'Column Filter' (with rule 'rm "final weight"'), 'Row Filter' (with rule 'born outside US'), and 'CSV Writer' (configured to 'write new file'). A yellow text box below the workflow states: 'This workflow is my first KNIME workflow. It reads data, removes uninteresting columns and rows from the data table and writes the result to a CSV file.' A red box labeled 'Node Description' points to the 'File Reader' node's configuration area.
- Node Description:** A pane on the top right showing the configuration for the 'File Reader' node. It explains that the node reads data from an ASCII file or URL and can be configured to read various formats. It also mentions a 'Server Workflow Project...' section with a dropdown menu, a 'Workflow Server' field set to 'publicserver.knime', and a 'Status: not connect' indicator with a 'Connect' button. A red box labeled 'KNIME Server' points to this section.
- Outline:** A small pane at the bottom left showing a thumbnail of the current workflow.
- Console:** A pane at the bottom right displaying the KNIME console output. It shows a welcome message, copyright information (2003-2011), and a warning: 'WARN CSV Writer Selected output file exists and will be overwritten!'. A red box labeled 'Console' points to this pane.



Many Available Data Nodes

Read

- File Reader
- ARFF Reader
- CSV Reader
- Line Reader
- Table Reader
- PMML Reader
- Model Reader
- Read XLS Sheet Names
- XLS Reader

Write

- CSV Writer
- ARFF Writer
- Table Writer
- PMML Writer
- Model Writer
- XLS Sheet Appender
- XLS Writer

Other

- Modular Data Generation
- Data Generator
- Create Table Structure
- Extract System Properties
- Extract Context Properties
- Image Column Writer
- Image Port Writer
- List Files
- Read PNG Images
- Table Creator

Many Available Community Nodes

- Community Nodes
 - KNIME Image Processing
 - CDK
 - DYMATRIX Customer Intelligence
 - EMBL-EBI
 - Groovy Scripting
 - HCS Tools
 - Matlab Scripting
 - NGS
 - Palladian
 - Python Scripting
 - R Scripting
 - RDKit
 - REST REST Nodes
 - Talete
 - Vernalis
 - CheS-Mapper

Many Statistics and Data Mining Nodes

Statistics

- Hypothesis Testing
 - Single sample t-test
 - Independent groups t-test
 - Paired t-test
 - One-way ANOVA
- Regression
 - Linear Regression Learner
 - Polynomial Regression Learner
 - Logistic Regression Learner
 - Regression Predictor
- Linear Correlation
- Correlation Filter
- Statistics
- Crosstab
- Value Counter

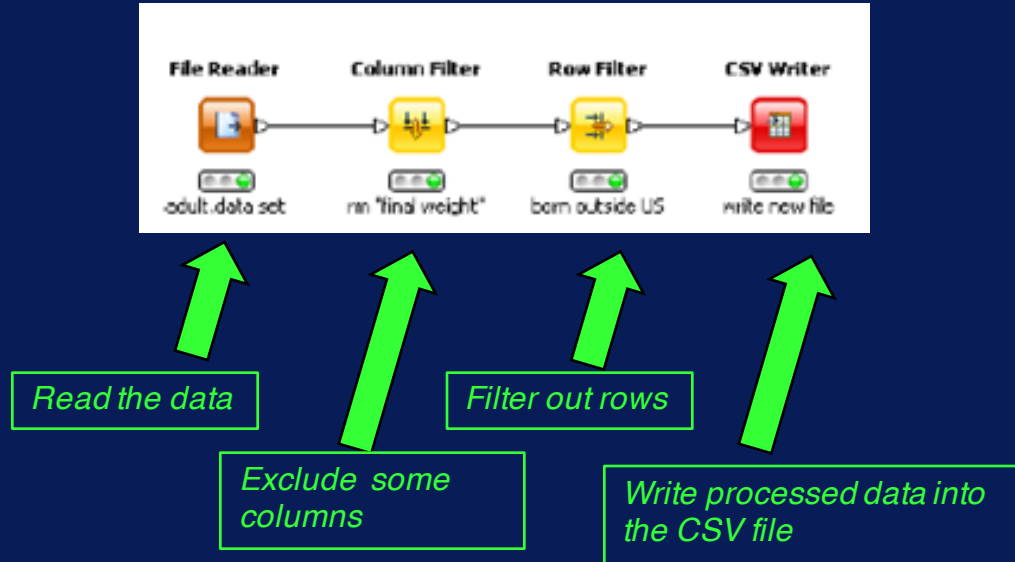
Mining

- Bayes
- Clustering
- Rule Induction
- Neural Network
- Decision Tree
 - Decision Tree Learner
 - Decision Tree Predictor
 - Decision Tree To Image
- Misc Classifiers
- Ensemble Learning
 - PMML Ensembles
 - Utility Nodes
- Bagging
- Boosting Learner
- Boosting Predictor
- Delegating

- Item Sets / Association Rules
- MDS
- PCA
- SVM
 - LIBSVM
 - SVM Learner
 - SVM Predictor
- Scoring
 - Enrichment Plotter
 - Entropy Scorer
 - Numeric Scorer
 - ROC Curve
 - Scorer
- Meta
 - Cross Validation
 - Feature Selection

Workflows in KNIME

- Sequence (pipeline) of the analysis steps
- Example
 - Read Data
 - Clean Data
 - Train a model
 - Etc.

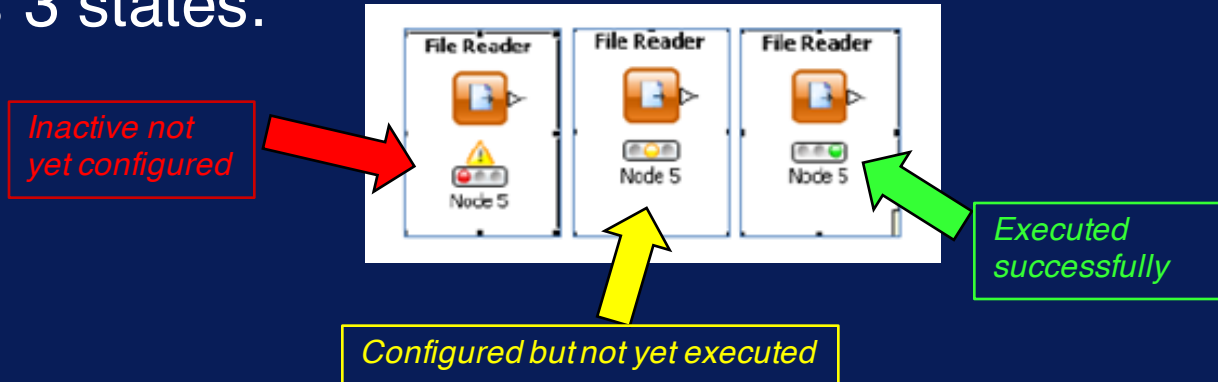


Building a Workflow

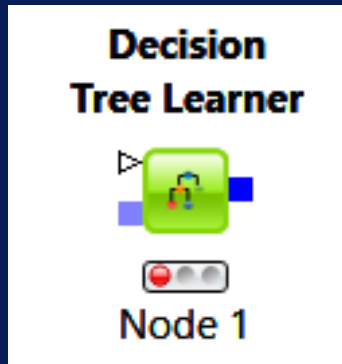
- A workflow is built by dragging nodes from the Node Repository to the Workflow Editor and connecting them
- Nodes are the basic processing units of a workflow
- Each node has a number of input- and/or output ports
- Data (or a model) is transferred via a connection from an out-port to the in-port of another node

Nodes in a Workflow

- Node is a single processing unit in the workflow
- The node takes data from the input, processes it and delivers it in the output port
- The processing can be anything from data manipulation, plotting to training Machine learning algorithms
- Every node has 3 states:



Decision Tree Node



Dialog - 0:1128 - Decision Tree Learner

File

Options PMMLSettings Flow Variables Memory Policy

General

Class column S Activity ▼

Quality measure Gini index ▼

Pruning method No pruning ▼

☒ Reduced Error Pruning

Min number records per node 2

Number records to store for view 10,000

☒ Average split point

Number threads 4

☒ Skip nominal columns without domain information

Binary nominal splits

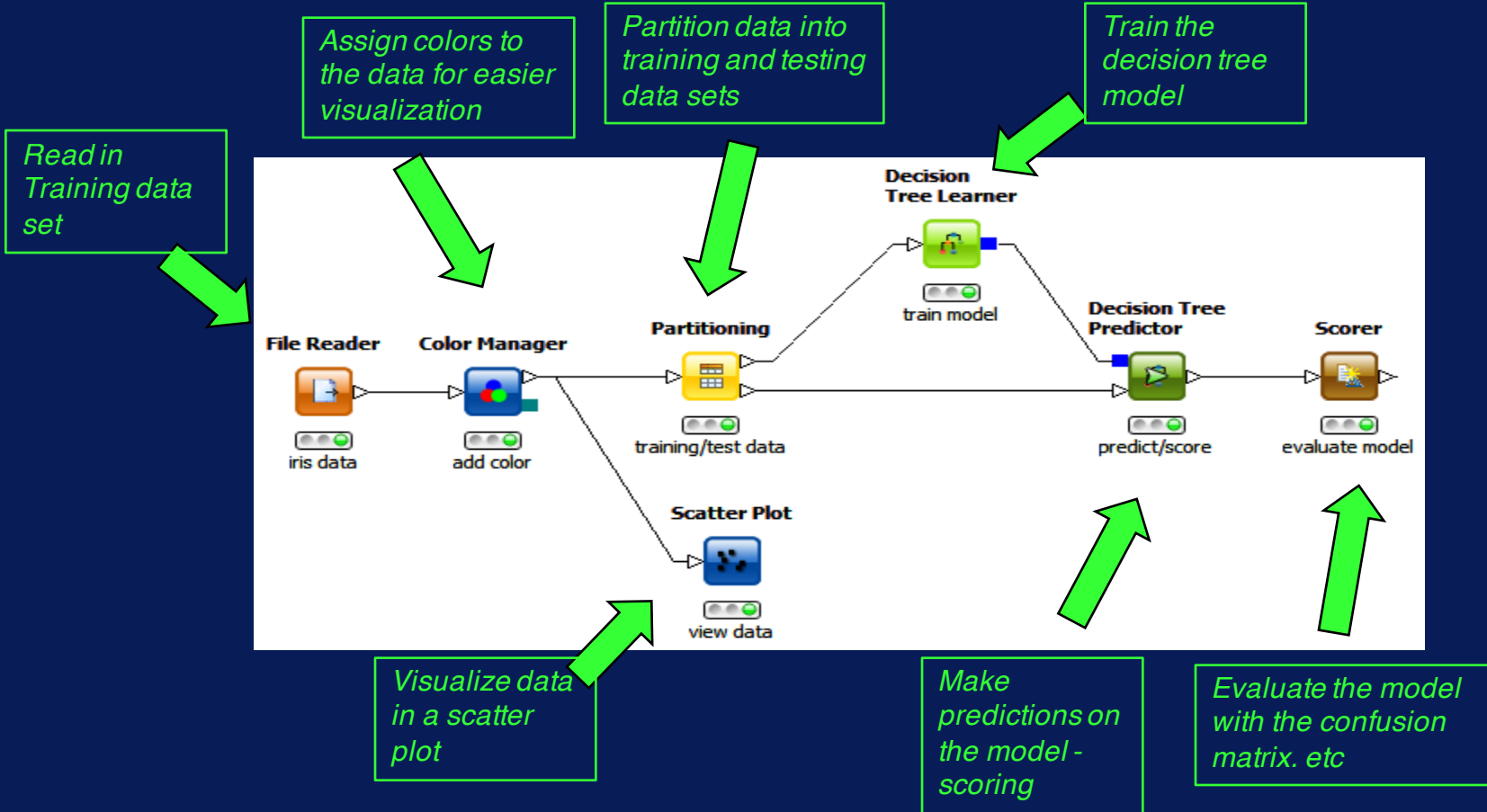
☐ Binary nominal splits

Max #nominal 10

☐ Filter invalid attribute values in child nodes

OK Apply Cancel ?

Decision Tree Workflow Example



To install a new KNIME extension:

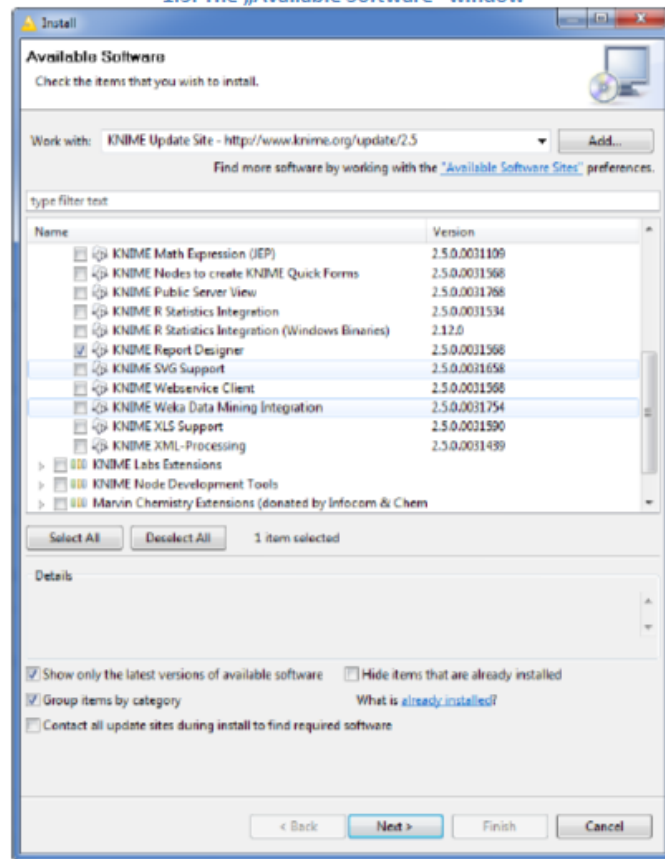
- From the Top Menu, select **"Help" -> "Install New Software"**
- In the "Available Software" window, in the "Work with" textbox, select the URL with the KNIME update site (usually named KNIME Update Site - <http://www.knime.org/update/2.x>)
- Select the extensions that you need. In this example, we chose to install the "KNIME Report Designer" in the category "KNIME & Extensions."
- Click the **"Next"** button on the bottom right and follow the wizard instructions

Once the KNIME extension has been installed and KNIME has been restarted, you should see a new node/category in the "Node Repository" in the KNIME workbench (see a few sections ahead).

For example, after installing the KNIME Report Designer, you should see a category "Reporting" in the "Node Repository" panel on the left.

Another way to reach the "Available Software" window and then install a new KNIME extension is the following: from the Top Menu, select **"File" -> "Install KNIME Extensions"**.

1.3. The „Available Software“ window



Additional Information

<https://www.knime.org/learning-hub?src=knimeapp>

Learning Hub

If you would like to learn more about KNIME, please check out our learning materials. We have made a large range of materials available on a variety of different media channels: books, whitepapers, the KNIME TV channel on Youtube, and of course the KNIME courses and webinars.

Usage

Applications

Development

Basic KNIME

KNIME Web site:

- Installation Guide
- KNIME Interactive Views
- List of KNIME Nodes and Features
- KNIME Extensions on Labs

Books:

- KNIME Press: *KNIME Beginner's Luck*
- PRACT Publishing: *KNIME Essentials*

KNIME TV on Youtube:

- Playlist "KNIME Starter Kit"
- Playlist "Basic KNIME"
- New Database Connectors (KNIME 2.10) (including DB Connectors to Hadoop Hive)
- New Database Nodes and Big Data Extension Nodes in KNIME 2.11
- Improvements in Excel nodes as part of the "What's New" talk at UGM 2014

Courses and Webinars:

- Basic User Training
- KNIME Online Self-Training

Advanced KNIME

Reporting

KNIME Server